One Model, Many Geometries: Spectral Dynamics of Multilingual LLMs

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1 Literature Review

We situate our project by first surveying approaches to multilingual training dynamics, introducing macro-level spectral metrics as a complementary methodology. Then, we review work connecting representation structure to cross-lingual performance, motivating our extension to the cross-lingual setting. Finally, we survey existing open-source multilingual LLMs to justify our setup.

Multilingual Training Dynamics. Several works explore the emergence of multilingual capabilities in LLMs using feature-level approaches. Riemenschneider and Frank (2025) and Wang et al. (2024) both trace neuron development over pretraining: the former finds that cross-lingual abstractions emerge over time, while the latter shows high correlation between neuron overlap and zero-shot cross-lingual transfer. Bayazit et al. (2026) takes a more mechanistic approach, training cross-coders across checkpoints to detect feature emergence and discontinuation.

Macro-Level Representation Geometry. Unlike feature-level approaches, macro-level spectral methods capture the global organization of the representation space. Their use is motivated by the anisotropy of LLMs' representation spaces (Ethayarajh, 2019), and Garrido et al. (2023) show that spectral metrics correlate with task accuracy. Li et al. (2025)–the direct motivation for our work–applies them to the study of pretraining dynamics. They use RankMe (Garrido et al., 2023)–a spectral metric of effective rank (Roy and Vetterli, 2007)–to identify three geometry phases during LLM pretraining: a warm-up compression, an entropy-seeking phase, and a final compression. However, their analysis is limited to English models and data, and it remains unclear whether other languages exhibit the same phases.

Representations and Cross-Lingual Transfer. Dubossarsky et al. (2020) show that spectral metrics of static monolingual embedding spaces predict cross-lingual task performance. However, Shah et al. (2024) find that geometric distances between language subspaces at a fixed checkpoint are inconsistent predictors of transfer performance. Blevins et al. (2022) take a dynamic approach, showing that transfer is asymmetric and acquired at different pretraining stages. Crucially, no work applies spectral methods over pretraining, leaving open whether the co-evolution of language geometries is a better indicator of cross-lingual performance than the geometry of specific checkpoints.

Open-Source Multilingual LLMs. Analyzing multilingual training dynamics requires models with public pretraining checkpoints. Prior work has used `BLOOM` (Workshop et al., 2023) (560M–7B parameters, 43 languages), but only 8 checkpoints are available. Models with more checkpoints include `Lucie` (Gouvert et al., 2025) (40) and the `Tri Series` (14), though both support fewer than 6 languages. We identify `Apertus-8B` (Apertus et al., 2025) (71 checkpoints, 15T tokens, 1800+ languages) and `FuxiTranyu-8B` (Sun et al., 2024) (58 checkpoints, 600B tokens, 40+ languages) as the most suitable models for their language diversity, checkpoint frequency and 8B size.

2 Contributions of Our Work

The main contribution of our work is studying **multilingual training dynamics** with **spectral methods**, directly extending Li et al. (2025). We further connect this analysis to downstream mono- and cross-lingual performance, investigating whether *differences* in geometry evolution predict *differences* in downstream mono- and cross-lingual performance. Finally, leveraging `Apertus-8B` and `FuxiTranyu-8B`, we conduct a fine-grained analysis of multilingual training dynamics.

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